

Critical-level Sufficientarianism

An axiomatic account of “enough”

Shuoxun ZHANG

Bossert, Cato & Kamaga (2022)

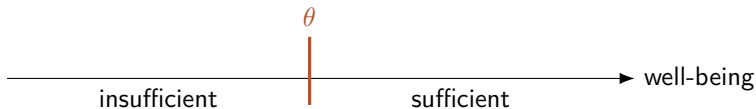
The sufficientarian question

*“What is important from the point of view of morality is not that everyone should have the **same** but that each should have **enough**.”*

— Frankfurt (1987)

Sufficientarianism

A general class of distributional principles that assign absolute priority to those below a threshold level that represents a minimally acceptable standard of well-being



Question: How should two distributions be ranked under sufficientarianism

From head counts to richer sufficientarianism

- ① **Frankfurt:** maximize the number of people at or above θ .
- ② With variable populations, one may instead compare the **proportion** above θ .
- ③ This is closely related to the head-count poverty index.

Problem

Head counts record whether someone crosses θ , but not **how far below** the threshold others remain.

Crossing the line $\neq$ measuring the depth

Two problems with a single threshold

Problem 1: depth of poverty

- Head counts incentivize helping people just below θ .
- The worst-off may receive no priority.

Crisp's response

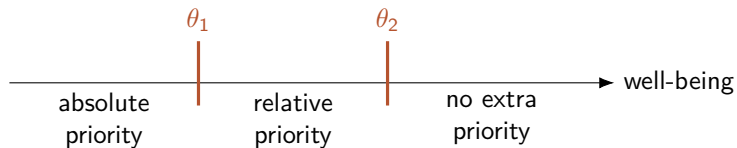
- Prioritarian weighting below θ .
- No special priority above θ .

Problem 2: excessive absolute priority

- Let $\theta = 10$.
- $u = (9, 100)$, $v = (9.1, 20)$.
- A single-threshold absolute-priority view ranks v above u : the insufficient person improves.

Casal's objection: a tiny gain below the threshold should not always dominate a huge loss above it.

Casal's two-threshold response

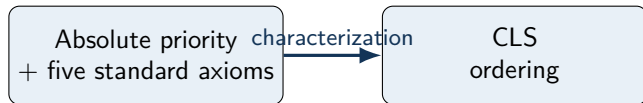


- Below θ_1 : absolute priority for severe poverty.
- Between θ_1 and θ_2 : favorable treatment, but trade-offs are permitted.
- Above θ_2 : no extra priority.

The main characterization result

Theorem 1

A sufficientarian ordering satisfies **absolute priority**, anonymity, strong Pareto, fixed-number independence, and continuity below and above θ **if and only if** it is a **critical-level sufficientarian (CLS)** ordering.



Setup and notation

- u_i : lifetime utility of person i ; utilities are interpersonally comparable and cardinal.
- $u = (u_1, \dots, u_n)$, $v = (v_1, \dots, v_m)$: distributions may have different population sizes.
- θ : a fixed sufficiency threshold.
- R : complete, transitive social-goodness ordering.

Notation	Meaning
$u R v$	u is at least as good as v
$u P v$	u is strictly better than v
$u I v$	u and v are equally good

- **Absolute priority.** For below-threshold u, v and above-threshold w, s ,

$$u P v \Rightarrow (u, w) P (v, s), \quad u I v \Rightarrow [(u, w) R (v, s) \Leftrightarrow w R s].$$

- **Anonymity.**

$$v = \pi(u) \Rightarrow v I u.$$

Individual labels do not affect social goodness.

- **Strong Pareto.**

$$[u_i \geq v_i \ \forall i, \ \exists i : u_i > v_i] \Rightarrow u P v.$$

- **Fixed-number independence.** For $u, v \in \mathbb{R}^n$ and $w, s \in \mathbb{R}^m$,

$$(u, w) R (v, w) \Leftrightarrow (u, s) R (v, s).$$

The number of unchanged people is fixed.

- **Continuity below θ .** If all utilities are at most θ and w is sufficiently close to u ,

$$u P v \Rightarrow w P v, \quad v P u \Rightarrow v P w.$$

- **Continuity above θ .** If all utilities are at least θ and w is sufficiently close to u ,

$$u P v \Rightarrow w P v, \quad v P u \Rightarrow v P w.$$

Critical-level sufficientarianism (CLS)

For a distribution $u \in \mathbb{R}^n$, let

$$L_n(u) = \{i : u_i < \theta\}, \quad H_n(u) = \{i : u_i > \theta\}, \quad E_n(u) = \{i : u_i = \theta\}.$$

Let g be a continuous, increasing transformation. R is a CLS ordering. For any $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^m$, $u R v$ if and only if either

$$\sum_{i \in L_n(u)} [g(u_i) - g(\theta)] > \sum_{i \in L_m(v)} [g(v_i) - g(\theta)],$$

or

$$\sum_{i \in L_n(u)} [g(u_i) - g(\theta)] = \sum_{i \in L_m(v)} [g(v_i) - g(\theta)],$$

$$\sum_{i \in H_n(u)} [g(u_i) - g(\theta)] \geq \sum_{i \in H_m(v)} [g(v_i) - g(\theta)].$$

A numerical illustration of the two stages

Let $\theta = 3$, with

$$g(-2) = -10, \quad g(-1) = -4, \quad g(1) = 2, \quad g(3) = 4, \\ g(6) = 6, \quad g(10) = 8, \quad g(22) = 12,$$

and consider

$$u = (1, 10, 1, 1, 6, 1), \quad v = (3, 10, -2, 10), \quad w = (22, -1).$$

	u	v	w
Below-threshold score $S_L(\cdot)$	-8	-14	-8
Above-threshold score $S_H(\cdot)$	6	—	8

- 1 Stage 1: $u P v$ and $w P v$, because $-8 > -14$.
- 2 Stage 2: $w P u$, because the below-threshold scores tie and $8 > 6$.

Population size is not itself the objective; the first concern is aggregate poverty below θ .

Why the axioms lead to CLS: proof intuition

- ① **Critical level.** Absolute priority + anonymity imply

$$(u, \theta) I u.$$

Adding a person exactly at the threshold is neutral.

- ② **Employing generalized-utilitarian on both side** On each side of θ , anonymity, Pareto, fixed-number independence, continuity, and this neutrality yield

$$\sum_i [g_{LE}(u_i) - g_{LE}(\theta)], \quad \sum_i [g_{HE}(u_i) - g_{HE}(\theta)].$$

- ③ **One function, lexicographic combination.** Normalize at θ and join the two pieces into g . Absolute priority says: first compare the lower-side sum; if tied, compare the upper-side sum.

Generalized-utilitarian sums on each side + absolute priority = CLS.
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Testing distributional sensitivity

Pigou–Dalton transfer principle

A transfer from a better-off person to a worse-off person is socially better if it does not reverse their relative ranks.

Let v be the pre-transfer distribution and u the post-transfer distribution:

$$u_i = v_i + \delta \leq v_j - \delta = u_j, \quad \delta > 0.$$



- i : recipient; j : donor.
- Everyone else is unchanged.

Testing distributional sensitivity

Transfer	Location condition	Requirement
Below threshold	$u_i = v_i + \delta \leq v_j - \delta = u_j < v_j \leq \theta$	$u P v$
Above threshold	$\theta \leq v_i < u_i = v_i + \delta \leq v_j - \delta = u_j$	$u P v$
Donor crosses down	$u_i = v_i + \delta \leq v_j - \delta = u_j \leq \theta < v_j$	$u P v$

A more equal distribution should be better.

When does CLS satisfy these transfer principles?

Testing distributional sensitivity

Strict concavity of g

$$g(\lambda x + (1 - \lambda)y) > \lambda g(x) + (1 - \lambda)g(y), \quad x \neq y, \quad 0 < \lambda < 1.$$

- Strict concavity means: **intermediate values beat extremes**.
- Thus, an equalizing transfer raises social value.

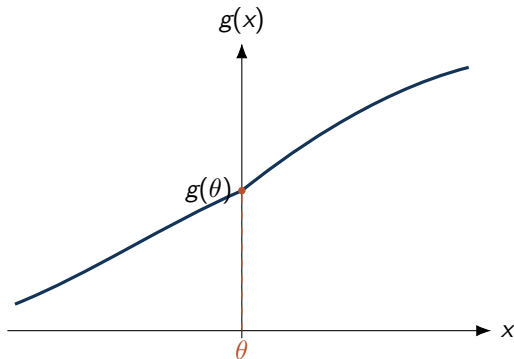
Theorem 2

The six axioms plus transfer principles below and above θ hold if and only if R is CLS and

$$g|_{(-\infty, \theta]} \quad \text{and} \quad g|_{[\theta, \infty)} \quad \text{are strictly concave.}$$

Global strict concavity of g is not required.

Testing distributional sensitivity



- g can be strictly concave on each side of θ .
- Its slope may jump upward at θ :

$$g'_-(\theta) < g'_+(\theta).$$

- Hence g need not be globally concave.

What about transfers across the threshold?

Testing distributional sensitivity

Recipient i remains below θ , while donor j crosses from above to below it:

$$v_i < v_i + \delta \leq v_j - \delta \leq \theta < v_j.$$

Before the transfer, Stage 1 counts only i ; afterwards, it counts both i and j . Therefore, $u P v$ if and only if

$$\underbrace{g(v_i + \delta) - g(\theta)}_{i \text{ after}} + \underbrace{g(v_j - \delta) - g(\theta)}_{j \text{ after}} > \underbrace{g(v_i) - g(\theta)}_{i \text{ before}}.$$

Equivalently,

$$\frac{g(v_i + \delta) + g(v_j - \delta)}{2} > \frac{g(v_i) + g(\theta)}{2}.$$

Testing distributional sensitivity

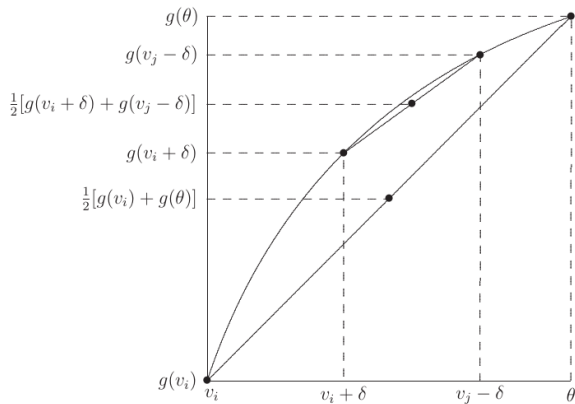


Figure 2. Across-threshold transfer principle

Repugnant conclusion and mere addition

- Adding a person below θ lowers the first-stage score. Thus CLS rejects the canonical repugnant conclusion when the added lives are below the sufficiency threshold.
- Let $\theta = 10$, $u = (1, 11, 11)$, and $v = (11, 11)$. Then

$$S_L(u) = -9 < 0 = S_L(v), \quad v P u.$$

A below-threshold addition is not neutral.

Sadistic-like implication with variable populations

- Since g is increasing, CLS does not prefer a negative utility to a positive one.
- But let $\theta = 10$ and $g(x) = x$:

$$A : 100 \text{ people at } 9, \quad S_L(A) = -100,$$

$$B : 1 \text{ person at } -1, \quad S_L(B) = -11.$$

$$B P A.$$

CLS avoids some population-ethics problems, but not all of their threshold-dependent variants.

Critical level = Threshold VS Critical $\neq$ Threshold